

第六章 求导的逆运算

6.1 原函数的概念

习题 6.1.1 求原函数

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| (1) $\int (2+x^5)^2 dx;$ | (2) $\int \left(\frac{1-x}{x}\right)^2 dx;$ |
| (3) $\int \left(1-\frac{1}{x^2}\right)\sqrt{x} dx;$ | (4) $\int \cosh x dx;$ |
| (5) $\int \sinh x dx;$ | (6) $\int \frac{2^{x+1}-5^{x-1}}{10^x} dx;$ |
| (7) $\int \frac{e^{3x}+1}{e^x+1} dx;$ | (8) $\int \sqrt{1-\sin 2x} dx;$ |
| (9) $\int \frac{dx}{(x+a)(x+b)};$ | (10) $\int \cos^2 x dx;$ |
| (11) $\int \sin^2 x dx;$ | (12) $\int \frac{x^5}{1+x} dx;$ |
| (13) $\int \frac{dx}{\cos^2 x \sin^2 x};$ | (14) $\int \frac{x^4}{1+x^2} dx.$ |

6.2 分部积分和换元法

习题 6.2.1 计算原函数

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| (1) $\int \arctan x dx;$ | (2) $\int \arcsin x dx;$ |
| (3) $\int x \operatorname{arccot} x dx;$ | (4) $\int x^2 \arccos x dx;$ |
| (5) $\int \ln(x+\sqrt{1+x^2}) dx;$ | (6) $\int \sqrt{x} \ln^2 x dx;$ |
| (7) $\int x^2 \cos x dx;$ | (8) $\int \frac{x}{\cos^2 x} dx;$ |
| (9) $\int \frac{\arctan x}{x^2} dx;$ | (10) $\int x^2 \cosh x dx;$ |
| (11) $\int \cos(\ln x) dx;$ | (12) $\int \sin(\ln x) dx.$ |

习题 6.2.2 设 p 是一个 n 次多项式, 试通过 p 的各阶导数来表示原函数

$$\int p(x)e^{ax} dx.$$

习题 6.2.3 求原函数:

(1) $\int x e^{-x^2} dx;$

(3) $\int \frac{\ln^2 x}{x} dx;$

(5) $\int \frac{dx}{\cosh x};$

(7) $\int \frac{dx}{\sin x};$

(9) $\int \frac{1}{x^2} \sin \frac{1}{x} dx;$

(11) $\int \frac{dx}{(a+bx)^2} \text{ (常数 } ab \neq 0);$

(13) $\int \cos ax \cos bx dx;$

(15) $\int \sin^4 x dx;$

(17) $\int \frac{\cos x + \sin x}{\sqrt[3]{\sin x - \cos x}} dx;$

(19) $\int \frac{\cos x \sin x}{a^2 \cos^2 x + b^2 \sin^2 x} dx;$

(21) $\int \frac{dx}{a \cos x + b \sin x};$

(23) $\int \frac{\ln x}{x \sqrt{1 + \ln x}} dx;$

(25) $\int \frac{x dx}{\sqrt{1+x^2+(1+x^2)^{3/2}}};$

(27) $\int e^{\sqrt{x}} dx;$

(29) $\int \arctan \sqrt{x} dx;$

(31) $\int \frac{x^{n/2}}{1+x^{n+2}} dx, (n \in \mathbf{N}^*);$

(33) $\int \frac{\cos x \sin x}{1 + \sin^4 x} dx;$

(2) $\int \frac{x}{\sqrt{1-x^2}} dx;$

(4) $\int \frac{x}{x^2+4} dx;$

(6) $\int \frac{dx}{\sqrt{x}(1+x)};$

(8) $\int \frac{dx}{x \ln x (\ln \ln x)};$

(10) $\int \frac{(\arctan x)^2}{1+x^2} dx;$

(12) $\int \cos ax \sin bx dx \text{ (} a, b \text{ 为常数)};$

(14) $\int \sin ax \sin bx dx;$

(16) $\int \sin^5 x dx;$

(18) $\int \frac{dx}{2 - \sin^2 x};$

(20) $\int \frac{dx}{\cos x + \sin x};$

(22) $\int \frac{\cos x}{\sqrt{2 + \cos 2x}} dx;$

(24) $\int \frac{x^3}{\sqrt{1-x^2}} dx;$

(26) $\int \frac{dx}{x \sqrt{x^2+1}};$

(28) $\int x^3 e^{-x^2} dx;$

(30) $\int x \sin \sqrt{x} dx;$

(32) $\int \frac{x^2}{\sqrt{x^2+a^2}} dx, (a \text{ 为常数});$

习题 6.2.4 推导出原函数

$$\int \ln^n x dx \quad (n \in \mathbf{N}^*)$$

的递推公式。

习题 6.2.5 用换元 $x = a \sinh t$ 或 $x = a \cosh t$ 解以下个题:

(1) $\int \frac{dx}{\sqrt{x^2+a^2}};$

(2) $\int \sqrt{x^2+a^2} dx;$

(3) $\int (x^2+a^2)^{-3/2} dx;$

(4) $\int \frac{x^2}{\sqrt{x^2-a^2}} dx;$

(5) $\int \sqrt{x^2-a^2} dx;$

习题 6.2.6 求原函数:

(1) $\int \left(1 - \frac{2}{x}\right)^2 e^x dx;$

(2) $\int \frac{x e^x}{(1+x)^2} dx.$

习题 6.2.7 求出函数 f , 设

(1) $f'(x^2) = \frac{1}{x} \quad (x > 0);$

(2) $f'(\sin^2 x) = \cos^2 x.$

6.3 有理函数的原函数

习题 6.3.1 求原函数:

(1) $\int \frac{x}{2x^2 - 3x - 2} dx;$

(2) $\int \frac{2x^2 + 1}{(x+3)(x-1)(x-4)} dx;$

(3) $\int \frac{dx}{2x^3 + 3x^2 + x};$

(4) $\int \frac{x^2}{x^3 + 5x^2 + 8x + 4} dx;$

(5) $\int \frac{x+3}{(x-2)^3} dx;$

(6) $\int \frac{x^2}{(x+2)^2(x+4)^2} dx;$

(7) $\int \frac{8x^3 + 7}{(x+1)(2x+1)^3} dx;$

(8) $\int \frac{dx}{(x+1)(x+2)^2(x+3)^3};$

(9) $\int \frac{dx}{1+x^3};$ 参考: 题7.7.1(6)和题20.4.3(3);

(10) $\int \frac{dx}{(x^2+9)^3};$

(11) $\int \frac{dx}{x(x^2+1)^2};$

(12) $\int \frac{dx}{1+x^4};$

(13) $\int \frac{1+x^2}{1+x^4} dx;$

(14) $\int \frac{dx}{x^4(1+x^2)}.$

解. (9). $\int \frac{dx}{1+x^3}$ 的解法:

方法 1: 通常解法 分解因式: $\frac{1}{1+x^3} = \frac{A}{1+x} + \frac{Bx+C}{x^2-x+1} = \frac{Ax^2-Ax+A+Bx+Bx^2+C+Cx}{1+x^3}$

因此存在关系: $A+C=1, A+B=0, -A+B+C=0$, 解得: $A=1/3, B=-1/3, C=2/3$. 即:

$$\frac{1}{1+x^3} = \frac{1}{3} \left(\frac{1}{1+x} + \frac{2-x}{x^2-x+1} \right).$$
 由此可得:

$$\begin{aligned} \int \frac{dx}{1+x^3} &= \frac{1}{3} \int \left(\frac{1}{1+x} + \frac{2-x}{x^2-x+1} \right) dx = \frac{1}{3} \int \left(\frac{1}{1+x} + \frac{1/2-x}{x^2-x+1} + \frac{3/2}{(x-1/2)^2+3/4} \right) dx \\ &= \frac{1}{3} \left(\int \frac{dx}{1+x} - \frac{1}{2} \int \frac{d(x^2-x+1)}{x^2-x+1} + \sqrt{3} \int \frac{d\left(\frac{2(x-1/2)}{\sqrt{3}}\right)}{\left(\frac{2(x-1/2)}{\sqrt{3}}\right)^2+1} \right) \\ &= \frac{1}{3} \ln|x+1| - \frac{1}{6} \ln|x^2-x+1| + \frac{\sqrt{3}}{3} \arctan\left(\frac{2x-1}{\sqrt{3}}\right) + c \end{aligned}$$

方法 2 做变形: $\int \frac{dx}{1+x^3} = \int \frac{1+x-x}{1+x^3} dx = \int \frac{dx}{1-x+x^2} - \int \frac{x}{1+x^3} dx$

令: $I = \int \frac{dx}{1+x^3}, J = \int \frac{x}{1+x^3} dx$, 则有:

$$I+J = \int \frac{dx}{1-x+x^2} = \int \frac{d(x-1/2)}{(x-1/2)^2+3/4} = \frac{2}{\sqrt{3}} \arctan\left(\frac{2x-1}{\sqrt{3}}\right) + c_1$$

同时有:

$$I-J = \int \frac{1-x}{1+x^3} dx = \int \frac{1-x+x^2-x^2}{1+x^3} dx = \int \frac{dx}{1+x} - \frac{1}{3} \int \frac{dx^3}{1+x^3} = \ln|x+1| - \frac{1}{3} \ln|x^3+1| + c_2$$

两式相加, 即得:

$$I = \int \frac{dx}{1+x^3} = \frac{1}{2} \ln|x+1| - \frac{1}{6} \ln|x^3+1| + \frac{1}{\sqrt{3}} \arctan\left(\frac{2x-1}{\sqrt{3}}\right) + c$$

□

一个在级数上的应用 求: $\sum_{n=0}^{\infty} (-1)^n \frac{1}{3n+1} = 1 - \frac{1}{4} + \frac{1}{7} - \frac{1}{10} + \frac{1}{13} - \dots$

该级数为 Leibniz 级数, 是收敛的。

构造幂级数 $S(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n+1}}{3n+1}$, 则其收敛半径 $R=1$, 即在 $(-1, 1)$ 中定义了一个和函数 $S(x)$.

已知当 $x=0$ 时, 有 $S(0)=0$; 当 $x=1$ 时, $S(1)$ 是收敛的. $S(x)$ 的表达式难以求出, 但有:

$$S'(x) = \sum_{n=0}^{\infty} (-1)^n x^{3n} = \sum_{n=0}^{\infty} (-x^3)^n = \frac{1}{1+x^3}$$

则有:

$$S(x) - S(0) = \int_0^x S'(t) dt = \int_0^x \frac{dt}{1+t^3}$$

即:

$$S(x) = \int_0^x \frac{dt}{1+t^3} \quad (-1 < x < 1)$$

由于级数在右端点处收敛, 由 Abel 第二定理, $S(x)$ 在 $x=1$ 处左连续. 则有:

$$S(1) = \lim_{x \rightarrow 1^-} S(x) = \int_0^1 \frac{dt}{1+t^3} = \frac{1}{3} \left(\ln 2 + \frac{\pi}{\sqrt{3}} \right).$$

6.4 可有理化函数的原函数

习题 6.4.1 求原函数:

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| (1) $\int \frac{\sin^3 x}{\cos^4 x} dx;$ | (2) $\int \frac{dx}{\cos^4 x \sin x};$ |
| (3) $\int \frac{dx}{2 \sin x - \cos x + 5};$ | (4) $\int \frac{\cos x \sin x}{\cos x + \sin x} dx;$ |
| (5) $\int \frac{\sin x}{1 + \cos x + \sin x} dx;$ | (6) $\int \frac{\sin^2 x}{1 + \sin^2 x} dx;$ |
| (7) $\int \frac{dx}{\cos^4 x + \sin^4 x};$ | (8) $\int \frac{dx}{\cos^4 x \sin^4 x};$ |
| (9) $\int \frac{\sin x}{\cos^3 x + \sin^3 x} dx;$ | (10) $\int \sqrt{\tan x} dx;$ |
| (11) $\int \frac{dx}{1 + \varepsilon \cos x}$ (常数 $\varepsilon > 1$). | |

习题 6.4.2 求原函数:

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| (1) $\int \frac{dx}{1 + \sqrt[3]{1+x}};$ | (2) $\int \frac{\sqrt[3]{x}}{x(\sqrt{x} + \sqrt[3]{x})} dx;$ |
| (3) $\int \frac{x}{\sqrt{x+1} + \sqrt[3]{x+1}} dx;$ | (4) $\int \frac{\sqrt{1+x^2}}{2+x^2} dx;$ |
| (5) $\int \frac{dx}{x\sqrt{1+x^2}};$ | (6) $\int \frac{x^3}{\sqrt{1+2x^2}} dx;$ |
| (7) $\int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} dx;$ | (8) $\int \frac{dx}{x\sqrt{a^2-x^2}}, x < a, x \neq 0;$ |
| (9) $\int \sqrt{\frac{1-x}{1+x}} \frac{dx}{x^2};$ | (10) $\int \frac{dx}{(x+a)^2(x+b)^3};$ |
| (11) $\int \frac{dx}{(x+a)^m(x+b)^n}, m, n \in \mathbf{N}^*$ | (12) $\int \frac{dx}{(1+x^n) \sqrt[n]{1+x^n}}, n \in \mathbf{N}^*.$ |